

TinyDSCNN-48 : Model Architecture & Data Flow

AI 231 (Deep Learning) Machine Exercise 2 — From-Scratch Tiny Voice Command Model

14,282 PARAMS

26-class baseline (~14.5k for 32-cl)

44.6 KB ONNX

Static INT8 quantized footprint

1.5 s WINDOW

16 kHz Mono (24,000 samples)

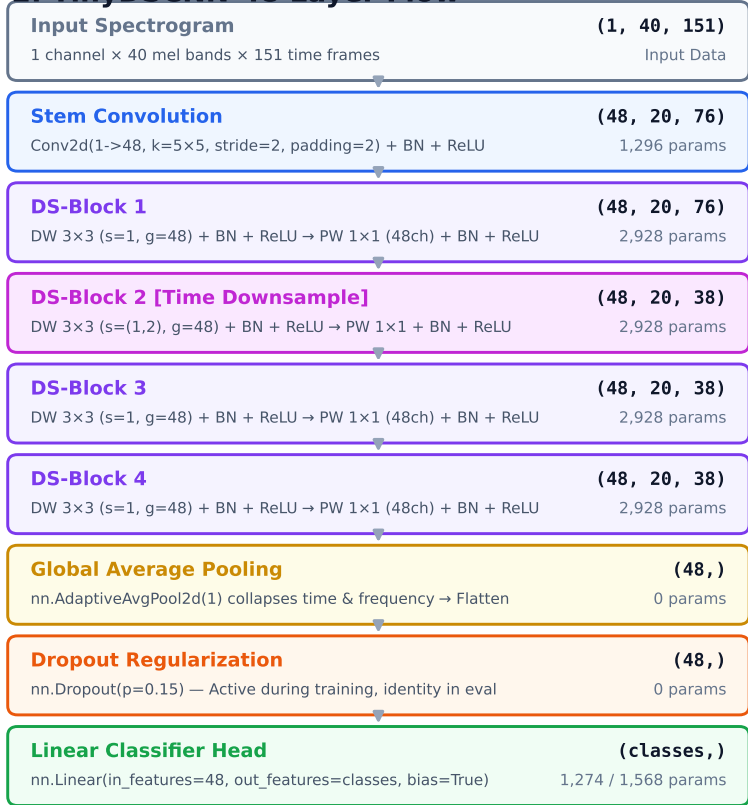
0.50 ms CPU

Single-thread x86 (< 10 ms RPi)

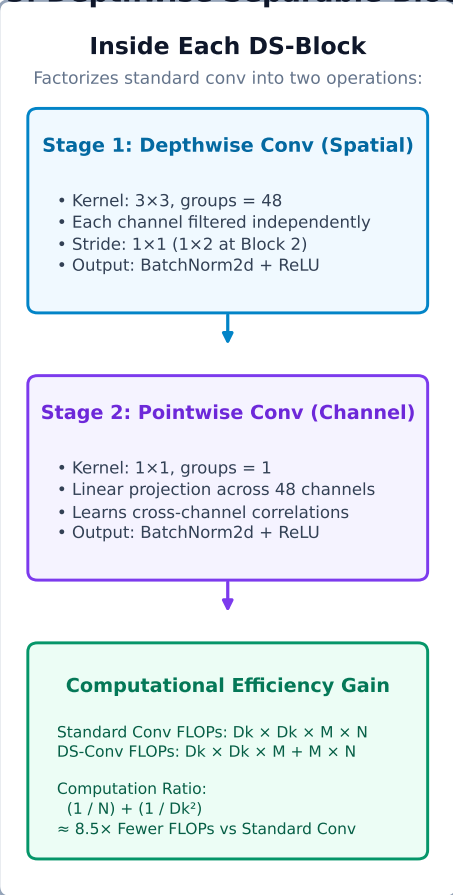
1. End-to-End Processing Pipeline



2. TinyDSCNN-48 Layer Flow



3. Depthwise Separable Block Detail



Designed for AI 231 ME2 | From-scratch initialization (Zero pretrained weights) | Exportable to static INT8 ONNX

TinyDSCNN-48 : Technical Specifications & Provenance

Layer Specifications, Computational Complexity, Edge Footprint, and Academic References

1. Layer-by-Layer Dimensions & Parameter Counts

Layer / Stage	Layer Operation / Class	Kernel / Stride / Config	Output Shape	Params
Input Audio	Log-Mel Spectrogram	16 kHz, 25ms win, 10ms hop	(1, 40, 151)	0
Stem Conv	nn.Conv2d + BN + ReLU	5×5, s=(2,2), p=2, bias=F	(48, 20, 76)	1,296
DS-Block 1 (DW)	nn.Conv2d (groups=48) + BN + ReLU	3×3, s=(1,1), p=1, bias=F	(48, 20, 76)	528
DS-Block 1 (PW)	nn.Conv2d + BN + ReLU	1×1, s=(1,1), p=0, bias=F	(48, 20, 76)	2,400
DS-Block 2 (DW)	nn.Conv2d (groups=48) + BN + ReLU	3×3, s=(1,2), p=1, bias=F [s_t=2]	(48, 20, 38)	528
DS-Block 2 (PW)	nn.Conv2d + BN + ReLU	1×1, s=(1,1), p=0, bias=F	(48, 20, 38)	2,400
DS-Block 3 (DW)	nn.Conv2d (groups=48) + BN + ReLU	3×3, s=(1,1), p=1, bias=F	(48, 20, 38)	528
DS-Block 3 (PW)	nn.Conv2d + BN + ReLU	1×1, s=(1,1), p=0, bias=F	(48, 20, 38)	2,400
DS-Block 4 (DW)	nn.Conv2d (groups=48) + BN + ReLU	3×3, s=(1,1), p=1, bias=F	(48, 20, 38)	528
DS-Block 4 (PW)	nn.Conv2d + BN + ReLU	1×1, s=(1,1), p=0, bias=F	(48, 20, 38)	2,400
Global Pooling	AdaptiveAvgPool2d(1) + Flatten	Global Average Pooling	(48,)	0
Regularization	nn.Dropout(p=0.15)	Active during training	(48,)	0
Classifier Head	nn.Linear(48 -> 26 classes)	Dense + Bias	(26,)	1,274
Total (26 classes)	Baseline Model (Synthetic Run)	100 CUDA epochs	Logits (26)	14,282
Total (32 classes)	Option B Dataset Adaptation	32 sub-classes / 19 intents	Logits (32)	14,576

2. Edge Deployment & Hardware Footprint

FP32 Uncompressed Model PyTorch .pt state_dict / FP32 ONNX weights	55.8 KB	INT8 Quantized ONNX Model Nine Conv2D layers and one Linear layer quantized to INT8	44.6 KB
Runtime RAM (RSS) Complete inference process including ONNX Runtime engine	73.45 MB	Single-Core CPU Latency P50 latency on standard x86 CPU thread; P95 is 0.679 ms	0.501 ms
Frontend + Inference Latency STFT + Mel filterbank (0.65 ms) + Neural forward pass (0.50 ms)	1.150 ms	Raspberry Pi Execution Easily satisfies real-time voice streaming constraint on ARM Cortex-A72/A76	< 10 ms

3. Academic References & Design Attribution

[1] Zhang, Y., Suda, N., Lai, L., & Chandra, V. (2017). "Hello Edge: Keyword Spotting on Microcontrollers." arXiv:1711.07128. ARM Applied ML Research.
[2] Howard, A. G., et al. (2017). "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications." arXiv:1704.04861. Google Inc.
[3] AI 231 ME2 Adaptation: Scaled to 48 channels, time-strided Block 2, and trained purely from-scratch with zero pretrained weights or foundation models.